

Assignment 3 – *Simple Shell*

Description:

This assignment was to build a simple shell in C that runs on top of the regular Linux terminal. The shell reads commands from the user, parses them, and executes them using fork and execvp. It also supports piping between commands using the | character.

Approach:

My first step was understanding what a shell actually does at a low level. I broke the problem into smaller pieces reading input, splitting it into tokens, running it as a process, and then adding pipe support at the end.

For reading input I used fgets with the 85 byte buffer limit the assignment requires. I checked for NULL return from fgets to handle both EOF and read errors separately since feof tells you which one it was.

For parsing I used strtok to split the input string on spaces and tabs into an array of strings ending with NULL, which is exactly what execvp needs.

For execution I used a fork to create a child process. The child calls execvp to replace itself with the actual command. The parent calls waitpid to wait for the child and then prints the child PID and exit status.

For pipes I split the input on the | character first to get individual command segments. I create all the pipes upfront before forking, then use dup2 to redirect each child's stdin and stdout through the correct pipe ends. The parent closes all pipe ends after forking so children can detect EOF properly.

Issues and Resolutions:

My first issue was a typo, I wrote char *token - strtok with a minus sign instead of equals sign. The compiler caught this and I fixed it.

My second issue was calling execvp with only one argument instead of two. execvp needs both the command name and the full args array. Once I added the second argument it worked correctly.

My third issue was forgetting to reset numCommands to 0 at the start of each loop iteration. This caused the second command entered to think there were leftover commands from before. I fixed it by adding numCommands = 0 before the strtok call inside the loop.

Analysis: Not required for this assignment

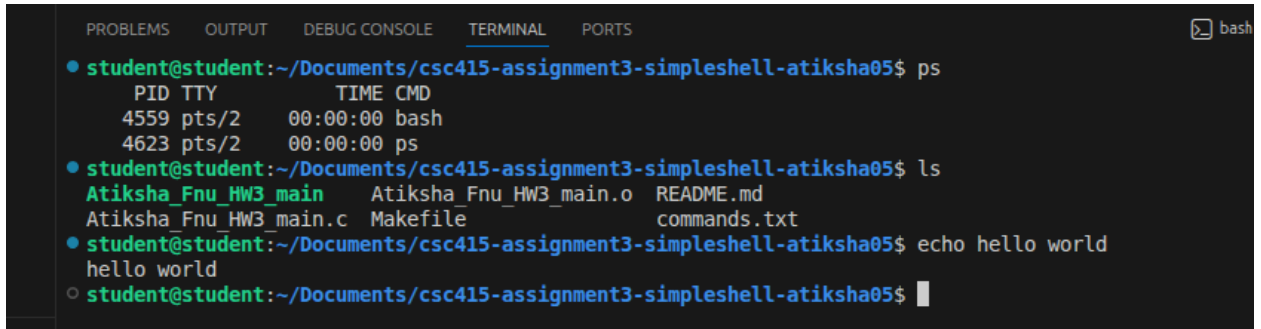
Screen shot of compilation:

```
student@student:~/Documents/csc415-assignment3-simpleshell-atiksha05$ make clean
rm *.o Atiksha_Fnu_HW3_main
student@student:~/Documents/csc415-assignment3-simpleshell-atiksha05$ make
gcc -c -o Atiksha_Fnu_HW3_main.o Atiksha_Fnu_HW3_main.c -g -I.
gcc -o Atiksha_Fnu_HW3_main Atiksha_Fnu_HW3_main.o -g -I. -l pthread
student@student:~/Documents/csc415-assignment3-simpleshell-atiksha05$
```

Screen shot(s) of the execution of the program:

```
student@student:~/Documents/csc415-assignment3-simpleshell-atiksha05$ make run
./Atiksha_Fnu_HW3_main "Prompt> "
Prompt> ps
  PID TTY          TIME CMD
 3933 pts/0    00:00:00 bash
 4426 pts/0    00:00:00 make
 4427 pts/0    00:00:00 sh
 4428 pts/0    00:00:00 Atiksha_Fnu_HW3
 4429 pts/0    00:00:00 ps
Child 4429 finished with status: 0
Prompt> ls
Atiksha_Fnu_HW3_main  Atiksha_Fnu_HW3_main.c  Atiksha_Fnu_HW3_main.o  Makefile  README.md  commands.txt
Child 4430 finished with status: 0
Prompt> ls -la
total 76
drwxrwxr-x 3 student student 4096 Feb 26 14:18 .
drwxr-xr-x 5 student student 4096 Feb 16 09:54 ..
drwxrwxr-x 7 student student 4096 Feb 16 09:54 .git
-rwxrwxr-x 1 student student 21968 Feb 26 14:18 Atiksha_Fnu_HW3_main
-rw-rw-r-- 1 student student 6725 Feb 26 13:53 Atiksha_Fnu_HW3_main.c
-rw-rw-r-- 1 student student 14576 Feb 26 14:18 Atiksha_Fnu_HW3_main.o
-rw-rw-r-- 1 student student 1864 Feb 16 09:57 Makefile
-rw-rw-r-- 1 student student 7444 Feb 16 09:54 README.md
-rw-rw-r-- 1 student student 66 Feb 16 09:54 commands.txt
Child 4431 finished with status: 0
Prompt> echo hello world
hello world
Child 4433 finished with status: 0
Prompt> cat /etc/hostname | wc -c
student
Child 4443 finished with status: 0
Prompt> exit
student@student:~/Documents/csc415-assignment3-simpleshell-atiksha05$
```

Regular terminal output for comparison:



The screenshot shows a terminal window with a dark background and light-colored text. At the top, there are tabs for 'PROBLEMS', 'OUTPUT', 'DEBUG CONSOLE', 'TERMINAL' (which is selected), and 'PORTS'. In the top right corner, there is a small icon of a terminal and the text 'bash'. The terminal content shows a series of commands and their outputs:

- A bullet point followed by the prompt `student@student:~/Documents/csc415-assignment3-simpleshell-atiksha05$` and the command `ps`. The output is a table with columns 'PID', 'TTY', 'TIME', and 'CMD'. It lists two processes: `4559 pts/2 00:00:00 bash` and `4623 pts/2 00:00:00 ps`.
- A bullet point followed by the prompt `student@student:~/Documents/csc415-assignment3-simpleshell-atiksha05$` and the command `ls`. The output lists files and directories: `Atiksha_Fnu_HW3_main`, `Atiksha_Fnu_HW3_main.o`, `README.md`, `Atiksha_Fnu_HW3_main.c`, `Makefile`, and `commands.txt`.
- A bullet point followed by the prompt `student@student:~/Documents/csc415-assignment3-simpleshell-atiksha05$` and the command `echo hello world`. The output is `hello world`.
- A circle followed by the prompt `student@student:~/Documents/csc415-assignment3-simpleshell-atiksha05$` and a cursor.